

Microcontroller Based Systems & Robotics Hub

Promoting robotic design and entrepreneurship experiences among students and teachers

Lesson 6:
Introduction to Motors

CONTENTS

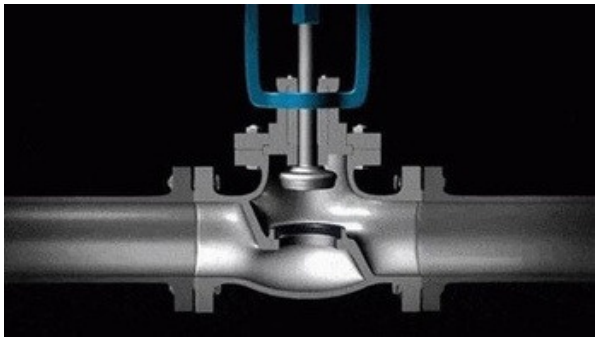


- DC motors (brushed and brushless)
- Pulse width modulation (PWM)
- Servo motors
- Motor control commands

- TASK/ACTIVITY: Motor control

ACTUATOR

- An actuator is a component thatis responsible for moving and controlling a mechanism or system
- Examples of systems that useactuators:



Valve actuators



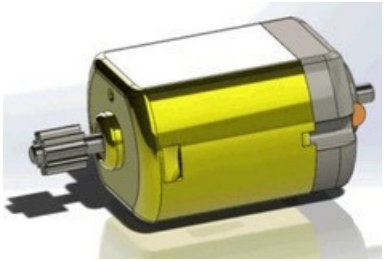
Electric motors



Pneumatic actuator

TYPES OF ACTUATORS

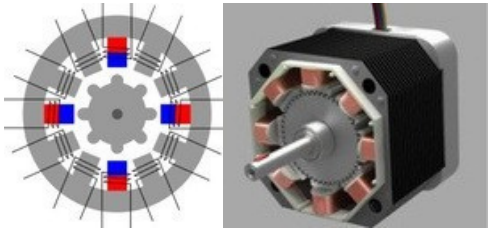
- DC motors
- Servo motors
- Stepper motor
- Hydraulics
- Pneumatic actuator



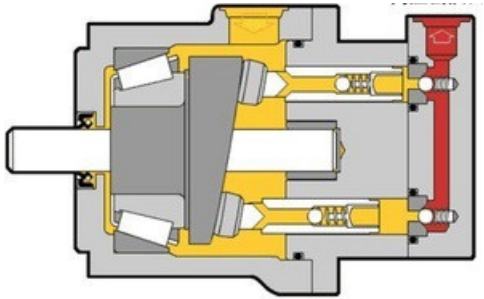
DC motor



Servo motor



Stepper motor



Hydraulic motor



Pneumatic cylinder

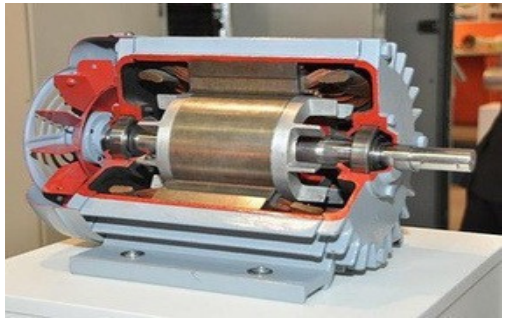
ELECTRIC MOTOR

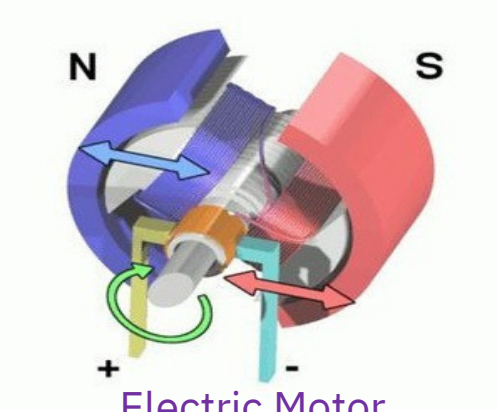
- Aelne cetlreictadtor is an machine that converts electrical energy into mechanical energy

Circuit symbol:

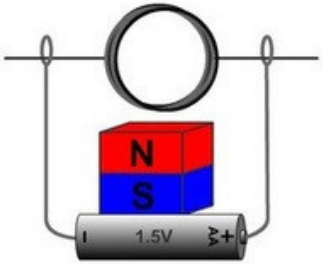


- An electric generator operates in the reverse direction, converting mechanical energy into electrical energy

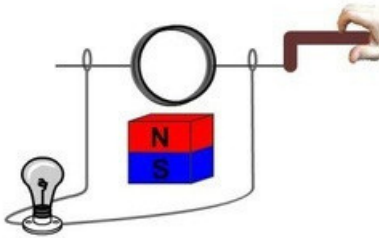




Electric Motor



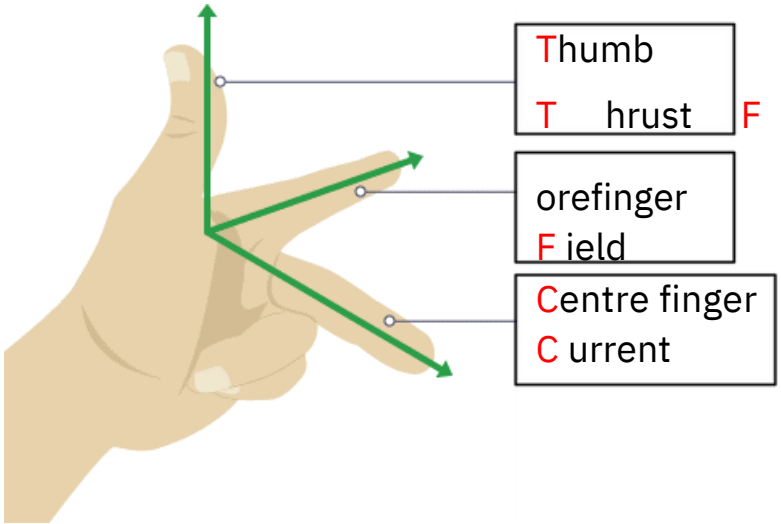
Motor



Generator

ELECTRIC MOTOR -WORKING PRINCIPLE

- Interaction between am agnetic field and a current carrying conductor produces a force (called “Lorentz force”)

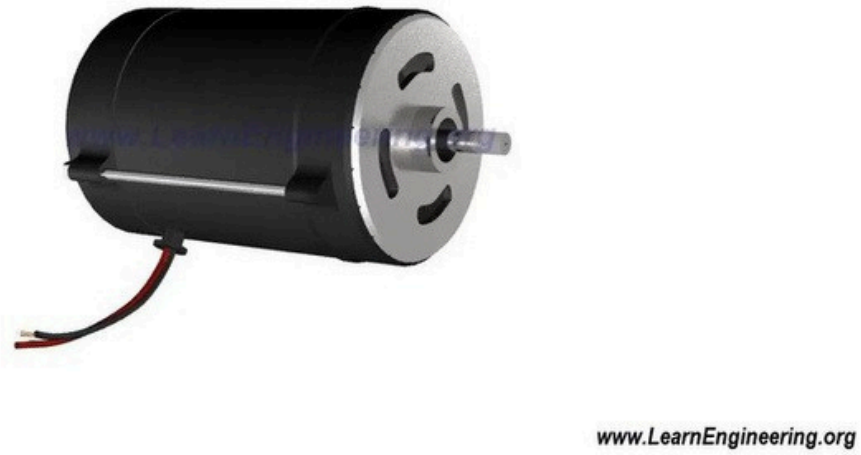


Source



Source

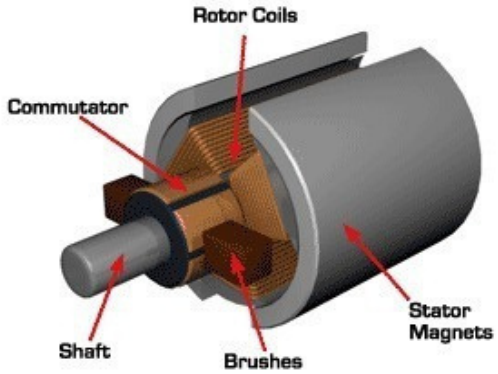
WORKING OF A DC MOTOR



Video

BRUSHED V/S BRUSHLESS DC MOTORS

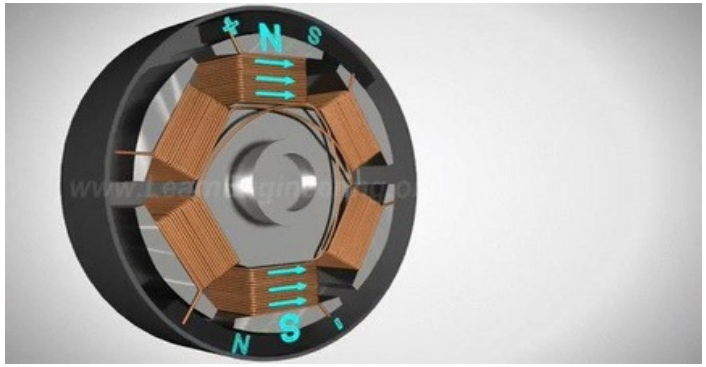
Brushed DC motor



- Permanent magnets for outer stator
- Rotating coils for inner rotor
- Commutator withmetal contact brushes reverse the polarity of the rotor
- May causesparkingdue to wear of brushes

to

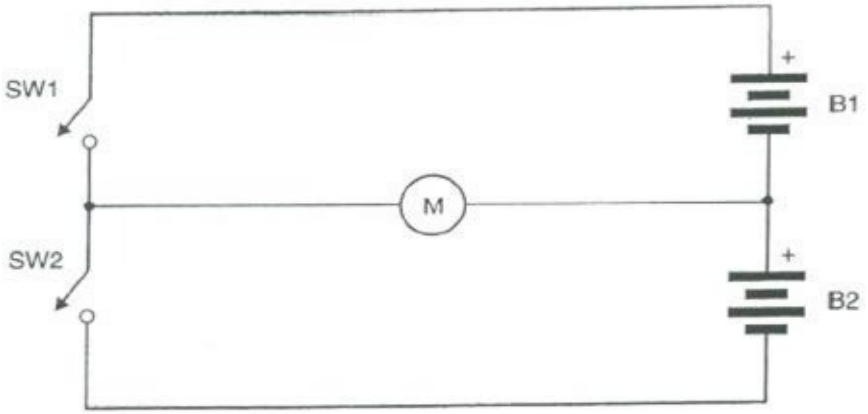
Brushless DC Motor, How it works ? -Lesics



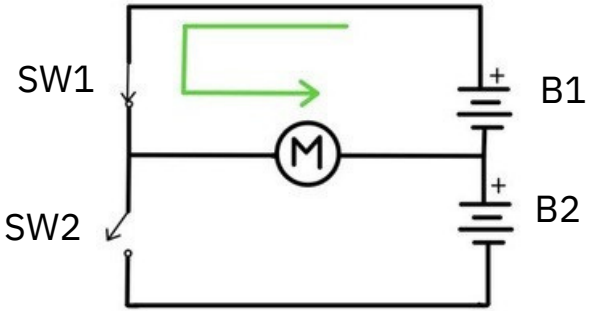
- Permanent magnets for outer rotor
- Rotating coils for inner stator
- No brushes
- No sparking , less noisy, longer life

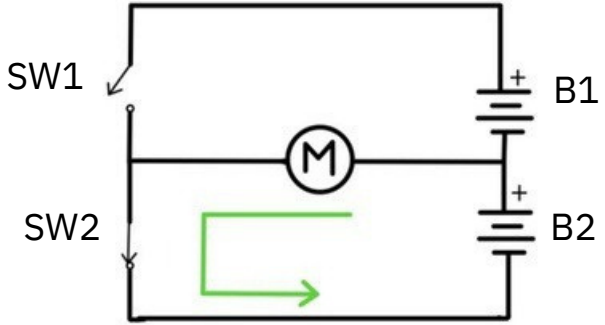
DC MOTOR DIRECTION CONTROL (Half-bridge)

- An H bridge is an electronic circuit that enables a voltage to be applied across a motor in the opposite directions
- Toexplain the H bridge, we begin with theconsideration ofahalf bridge
- Consider the circuit below with 2 voltage sources, 2switches andaDC motor:



HALF-BRIDGE CIRCUIT (using switches)





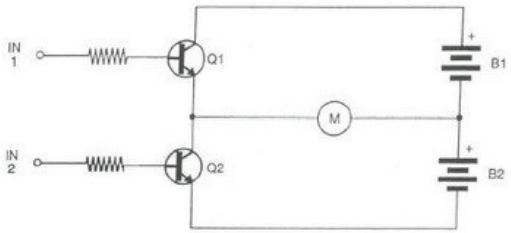
- When SW1 is closed, B1 is connected to the motor, current flows from left to right, motor
- turns in one direction
- When SW2 is closed, B2 is connected to the motor, current flows from right to left, motor turns in the opposite direction

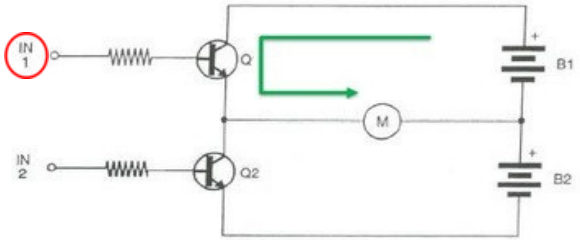
SW1 and SW2 cannot be closed simultaneously , as this leads to B1 and B2 being in short-circuit

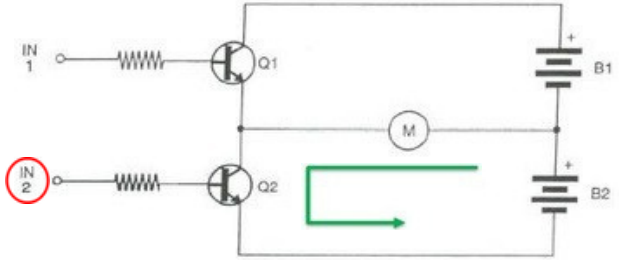
What electronic component can be used instead of a mechanical switch?

HALF-BRIDGE CIRCUIT (using transistors)

- Consider the circuit below with switches replaced by two NPN transistors for electrical switching:







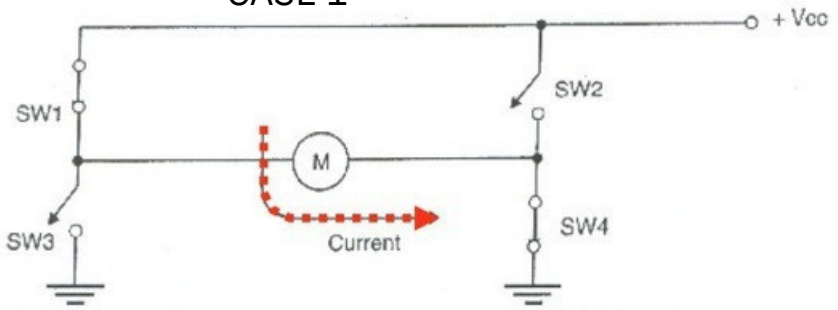
- CASE 1: IN1 is high, Q1 conducts → motor turns inforwarddirection byB1
- CASE 2: IN2 is high, Q2 conducts → motor turns in reverse direction byB2
- IN1 and IN2 cannot be driven high simultaneously , as this leads to B1 and B2 being in short-circuit

The main disadvantageof a half bridge DC motor drive circuit is that it requires a dual power supply

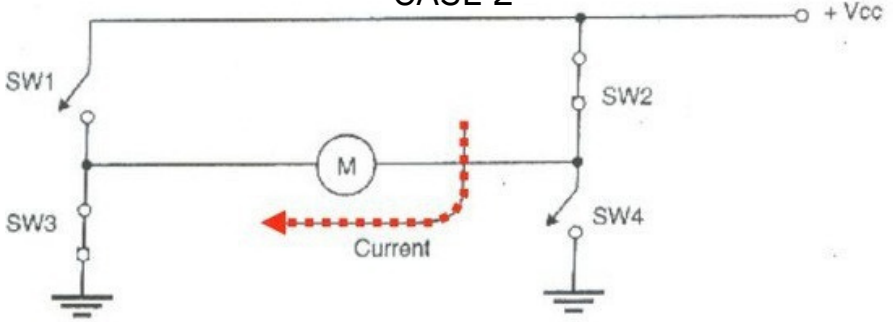
NOTE: These circuit diagrams are for conceptual understanding only, diodes will be required in the half-bridge circuit for control of a DC motor in real life

DC MOTOR DIRECTION CONTROL (Full-bridge)

CASE 1



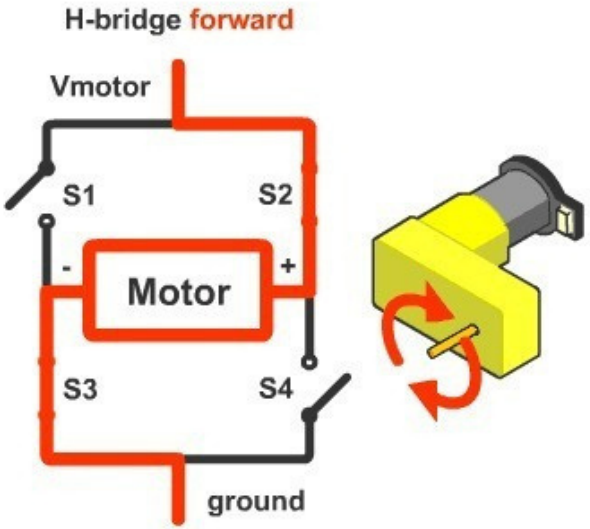
CASE 2



- A full-bridge circuitis calleda H-bridge (the shape of the circuitresembles the letter “H”)
- Considerthe circuitshow, which consistsof 4 switches, 1 voltagesource, and aDCmotor; forw ard re verse
  - CASE 1: SW1 and SW4 are closed (SW2 and SW3 open) → VCC drives motor in direction
  - CASE 2: SW2 and SW3 are closed (SW1 and SW4 open) → VCC drives motor in direction

NOTE: These circuit diagrams are for conceptual understanding only, diodes will be required in the H-bridge circuit for control of a DC motor in real-life

H-BRIDGE CIRCUIT (using switches)

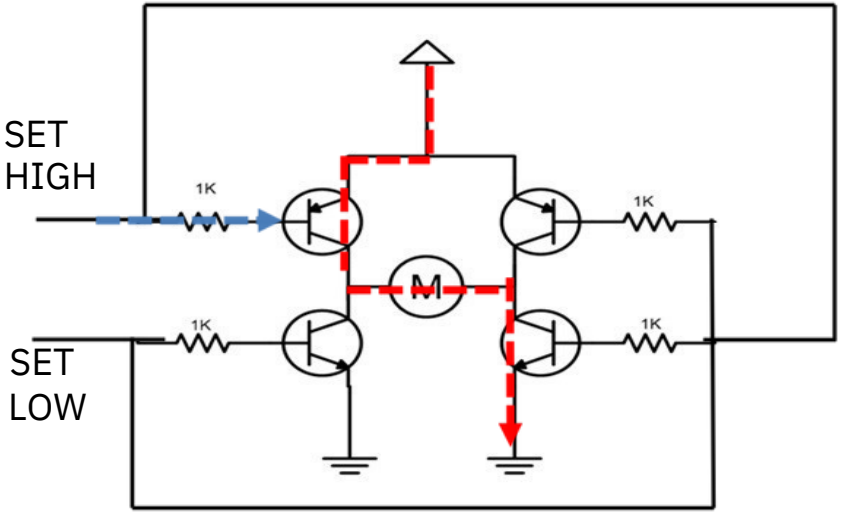


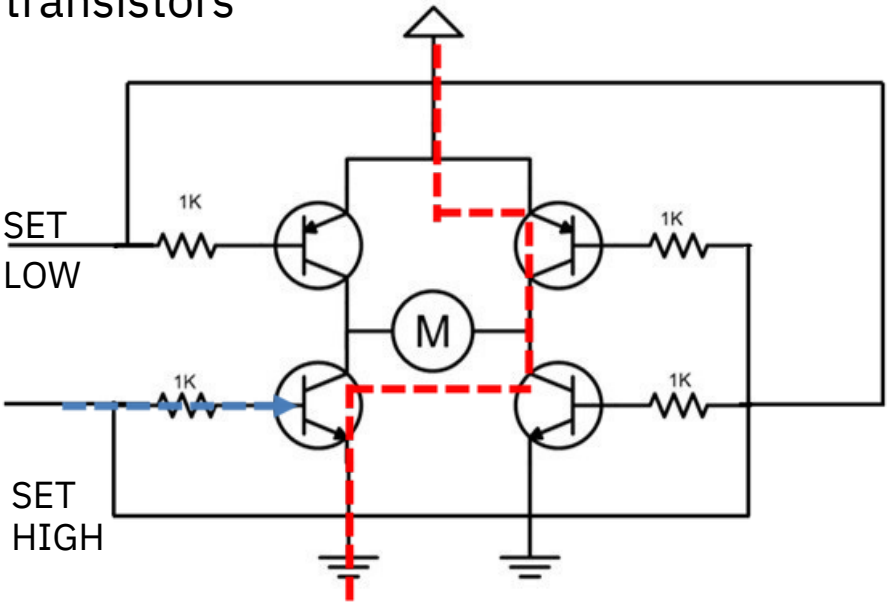
H-Bridge

NOTE: These circuit diagrams are for conceptual understanding only, diodes will be required in the H-bridge circuit for control of a DC motor in real life

H-BRIDGE CIRCUIT (using transistors)

H-bridge using transistors

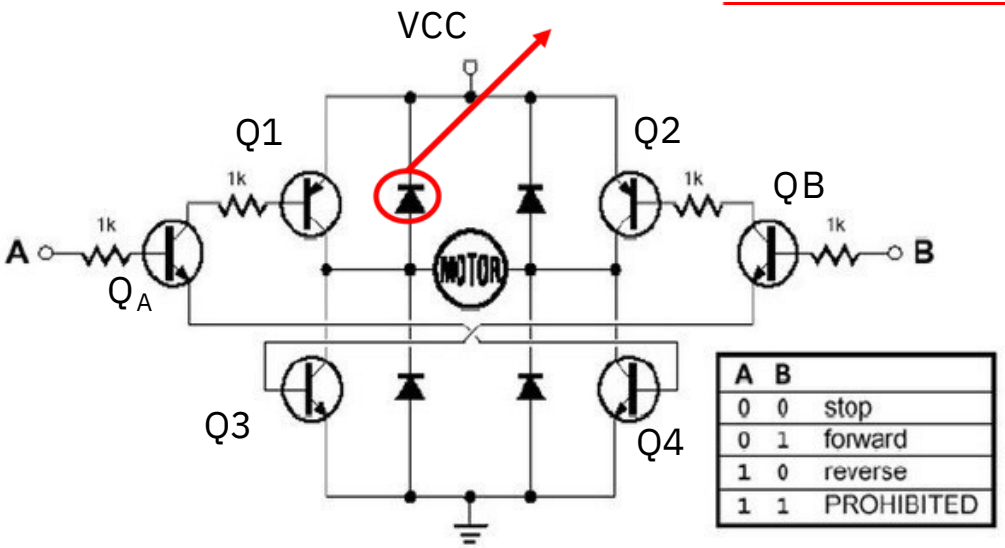




NOTE: These circuit diagrams are for conceptual understanding only, diodes will be required in the H-bridge circuit for control of a DC motor in real life

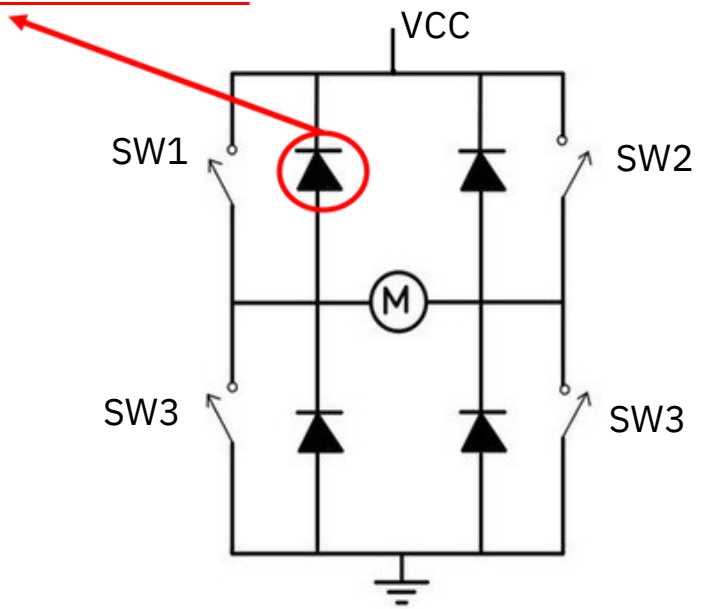
DC MOTOR DIRECTION CONTROL (H-bridge)

Diodesare used to prevent damage to switching elements fromi nductive kickback



A	B	
0	0	stop
0	1	forward
1	0	reverse
1	1	PROHIBITED

(1) Using Transistors

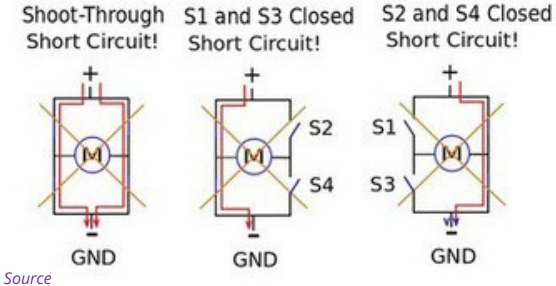


(2) Using switches

To control a DC motor in real-life, thefull-fledged H-Bridge circuit in figure (1) can beused

WHY IS DIODE REQUIRED IN A H-BRIDGE?

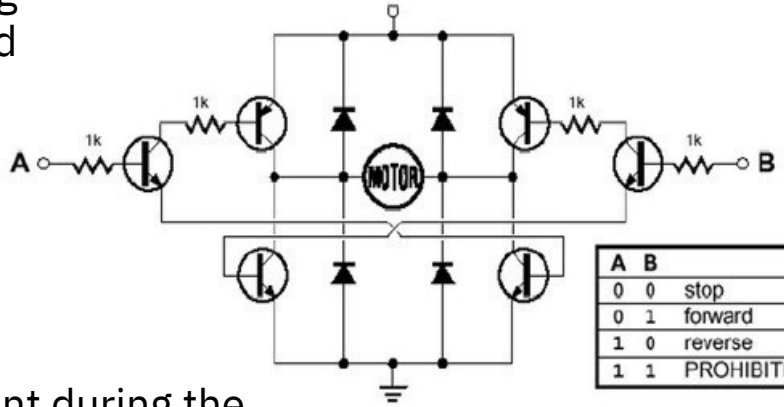
- In the prohibited case, i.e., turning ON switching elements on same side of the bridge simultaneously, creates a short circuit



Source

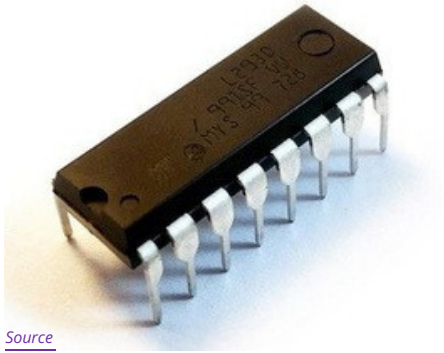
- While in other cases, when current is instantaneously cut off in a H-bridge (when switching from forward to reverse or vice-versa), the stored magnetic energy results in a high spike in voltageacross the motor (inductor):

V = L $\frac{di}{dt}$, V increases as the $\frac{di}{dt}$ term increases, which expressing a high flyback voltage, damages the transistors

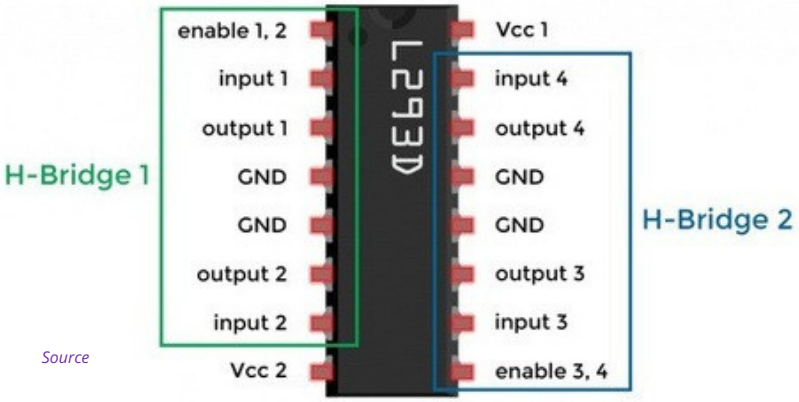


- In either case, the diodes provide a path for the current during the switching periods, dissipatingthe energy as heat, to protect the switchingelements.

DC MOTOR DIRECTION CONTROL (L293D IC)



Source

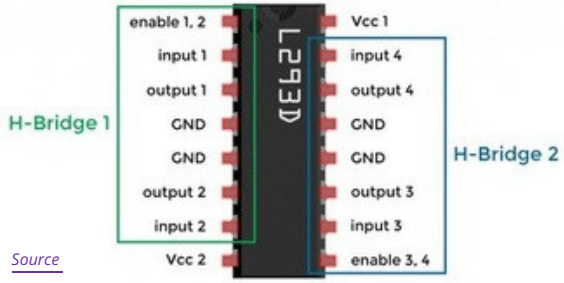


Source

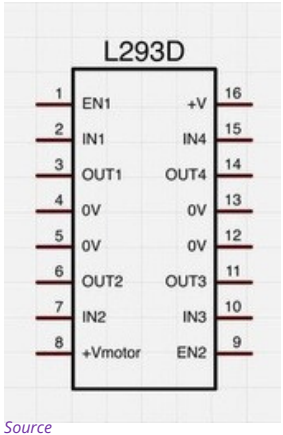
- Motor driver ICsare integratedcircuit chipsthat simplify controlof motors
- The L293D is a 16-pin Motor Driver IC which can control two DC motors simultaneously and independently
- Itcanprovide 600mA per channel at asupply voltage range of4.5V to 36V
- It has an internally embedded diode so there is no need of external diodes for interfacing the DC motor

ACTIVITY -1 (DC MOTOR: DIRECTION)

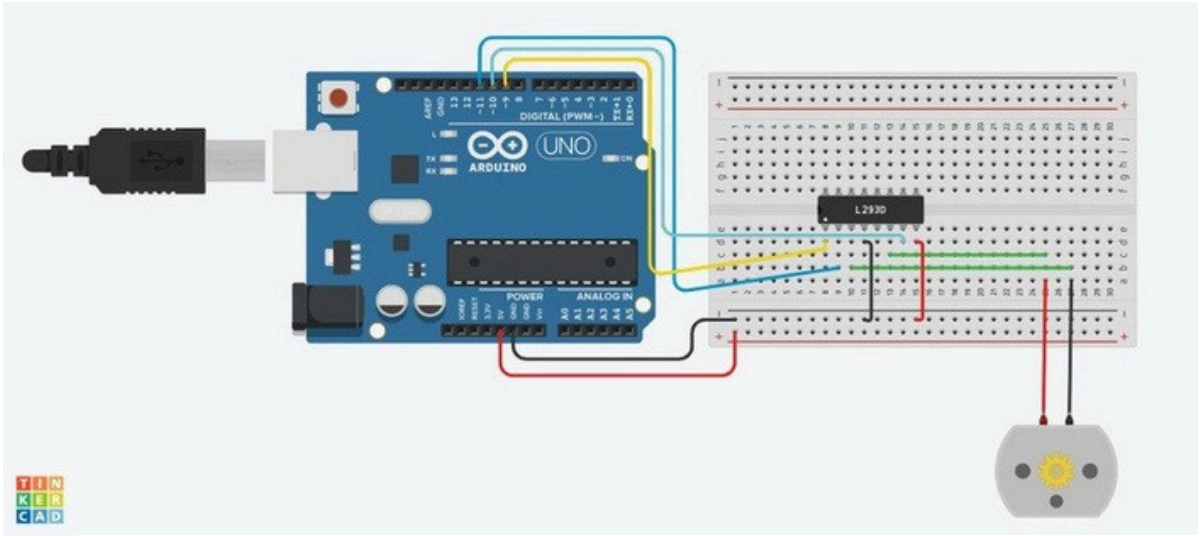
Wiring Arduino with L293D and DC Motor



Source



Source



ACTIVITY -1 (DC MOTOR: DIRECTION)

CODE

```
int IN1 = 11; // input 1
int IN2 = 10; // input 2
int EN1 = 9;  // enable pin

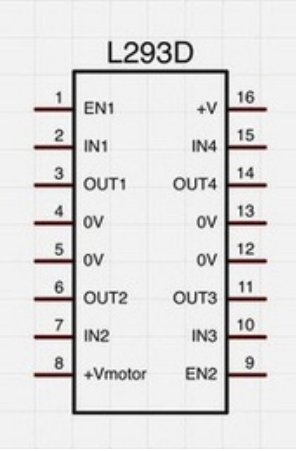
void setup() {
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(EN1, OUTPUT);
  // set IN1, IN2 and EN1 pins in OUTPUT mode
  digitalWrite(EN1, HIGH);
  // set enable pin on L293D HIGH
}
```

Program

```
void loop() {

  digitalWrite(IN1, LOW);
  // set pin 2 on L293D LOW
  digitalWrite(IN2, HIGH); //CW
  // set pin 7 on L293D HIGH, turn CW
  delay(3000);
  // for 3 seconds

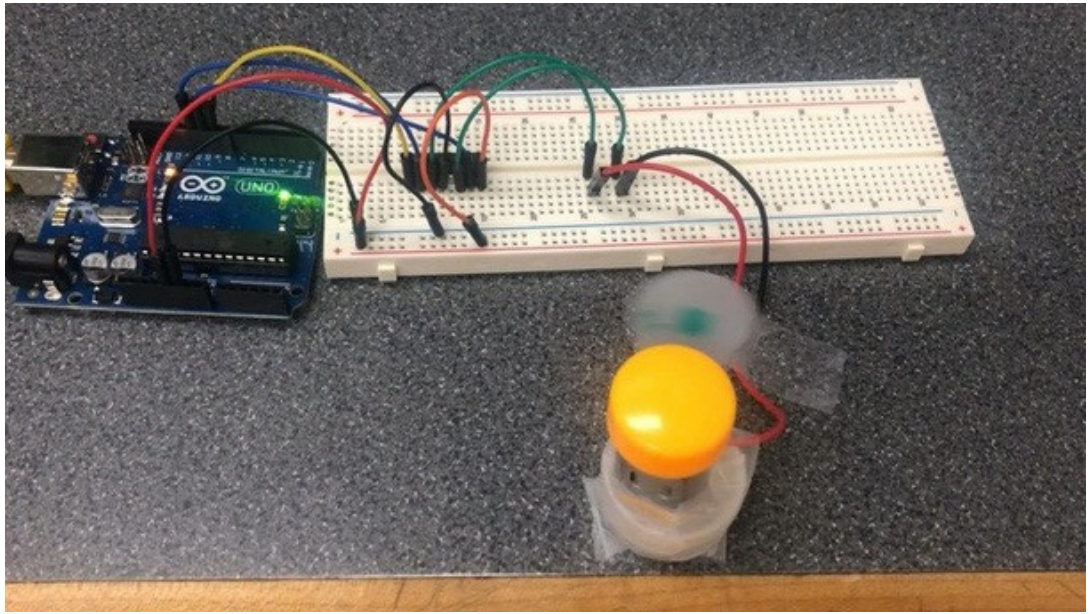
  digitalWrite(IN1, HIGH);
  // set pin 2 on L293D HIGH, turn CCW
  digitalWrite(IN2, LOW);
  // set pin 7 on L293D LOW
  delay(3000);
  // for 3 seconds
}
```



Source

ACTIVITY -1 (DC MOTOR: DIRECTION)

NOTE : Speed of the DC motor was reduced to clearly show the change in direction in the video

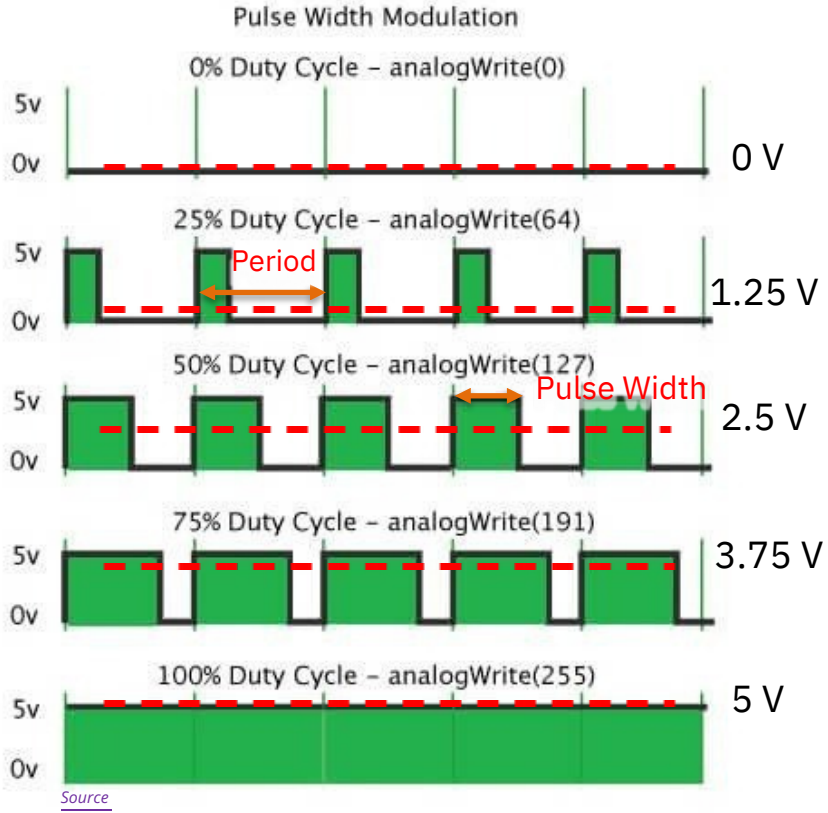


Video

PULSE WIDTH MODULATION

Effective voltage

DC motor speed can be controlled using PWM



Source

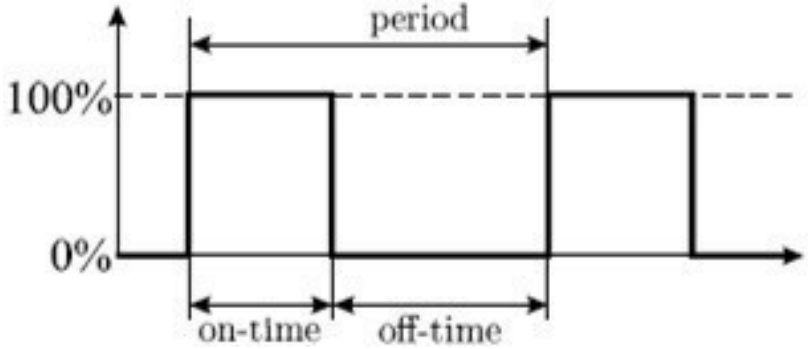
DUTY CYCLE

$$\text{duty cycle} = \frac{t_{ON}}{t_{ON} + t_{OFF}}$$

t_{ON} = ON time

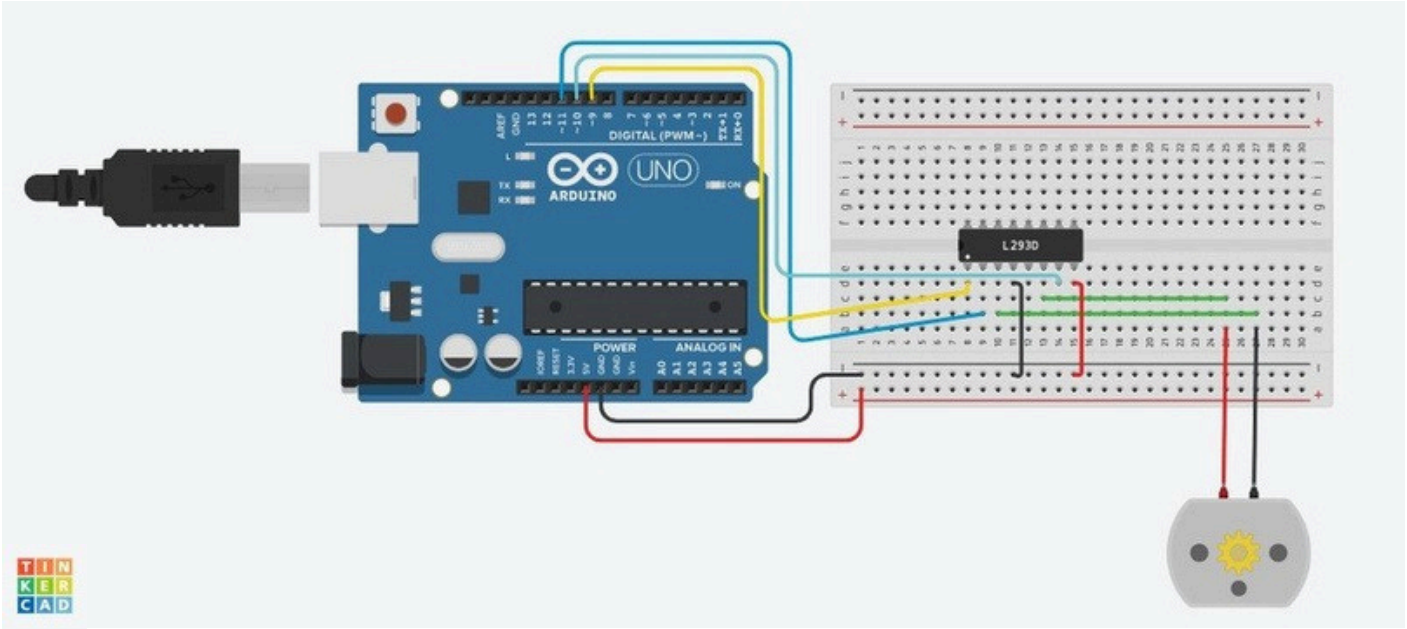
t_{OFF} = OFF time

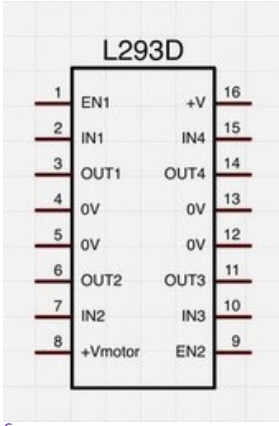
$t_{ON} + t_{OFF}$ = Time period



ACTIVITY -2 (DC MOTOR: SPEED)

Same circuit as ACTIVITY -1





Source

NOTE: Arduino pins with ”~” sign next to the pin number are PWM pins, used to control actuators using PWM (here, pins 10 & 11 is used to control a DC motor)

ACTIVITY -2 (DC MOTOR: SPEED)

DC Motor Speed Control Code

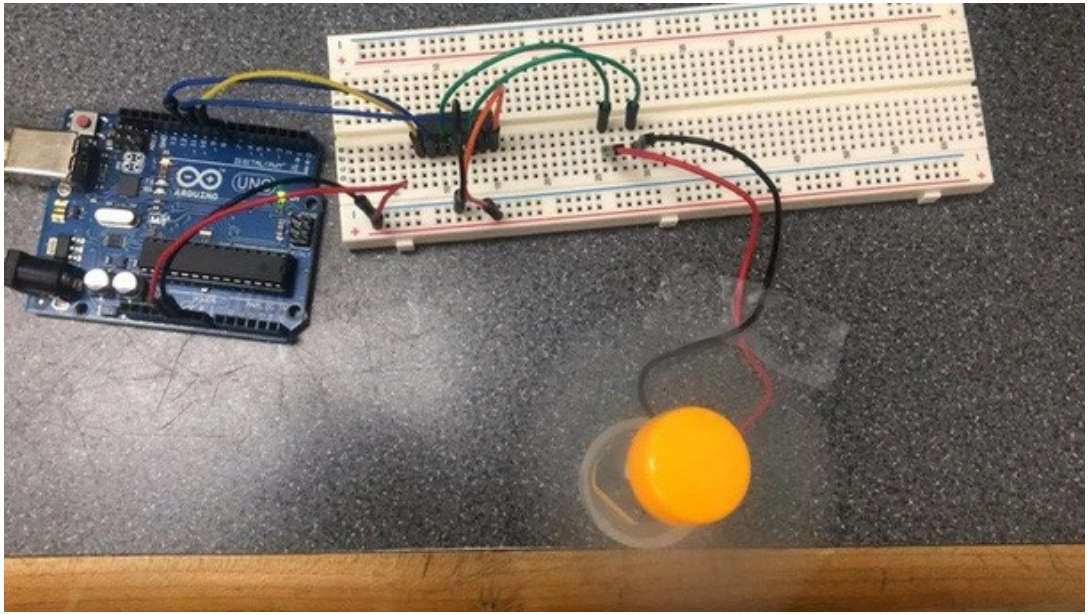
```
#define E1 9 // Enable Pin
#define IN1 11 // Control pin 1 for motor -- CW
#define IN2 10 // Control pin 2 for motor -- CCW

void setup()
{
  pinMode(E1, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
}
```

```
void loop()
{
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  analogWrite(E1, 200);
  // value between 0-255 Enable pin controls PWM
  // 200 --> 3.9V (duty cycle = 200/255 ~ 78%)
  delay(4000);
  analogWrite(E1, 110);
  // reduce the speed by about half (2.15V)
  delay(4000);
  analogWrite(E1, 85);
  // reduce the seed further (1.67V)
  delay(4000);
}
```

Program

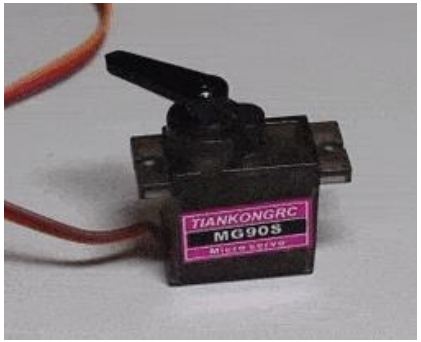
ACTIVITY -2 (DC MOTOR: SPEED)



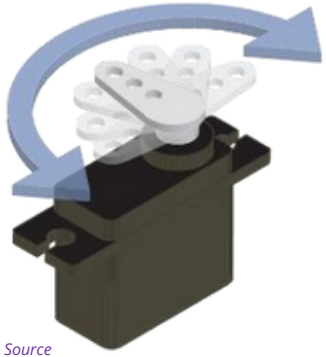
Video

SERVO

- Servo motoris a type of actuator used for angular positioning
- Standard servotypically has a movement range of 180 degrees
- Continuous servo has a freedom to complete one full rotation



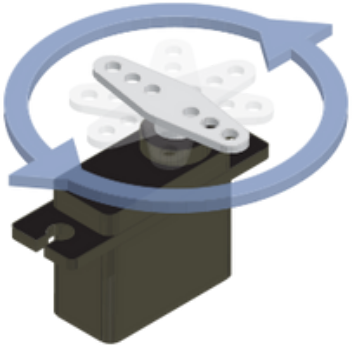
Standard Servo motor



Source



Continuous Servo Motor



Source

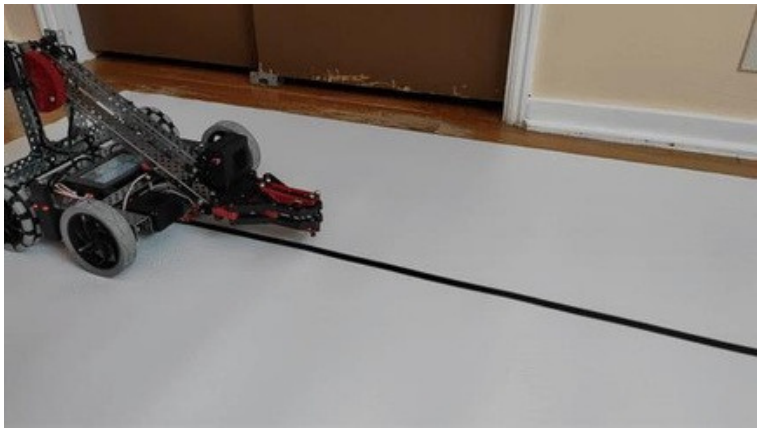
STANDARD V/S CONTINUOUS SERVO

Sontalyn dtuarnds s oevrveor a rangeContinuous rotationser voturns (usually 0°-180°), with precise feedbackcontinuously, withcontrolover its control over its angular positionspeed and direction



Source

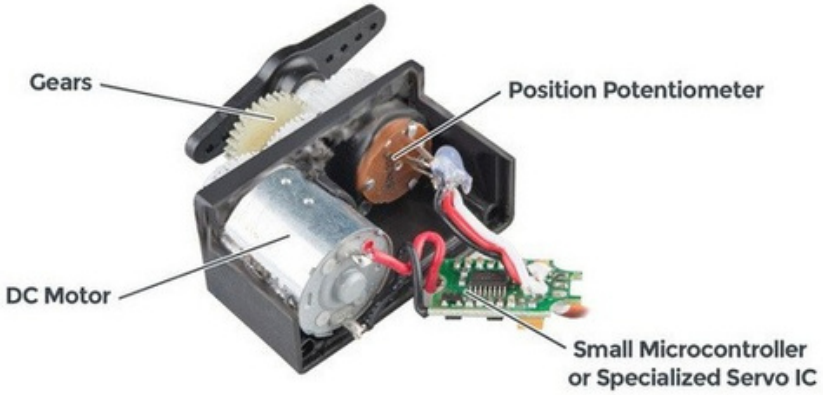
Standard servo example: Robotic arm



Source

Continuous servo example: Mobile Robot

SERVO: SECTIONAL & EXPLODED VIEW

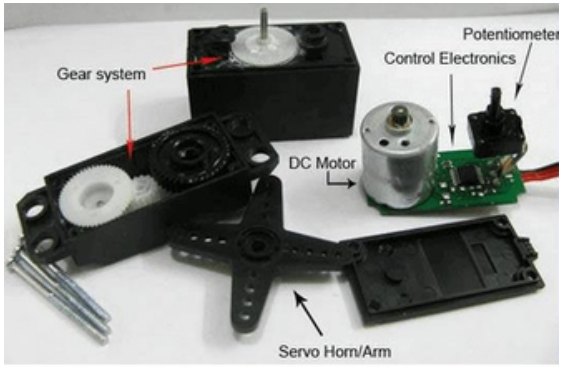


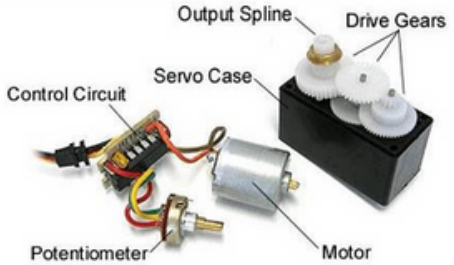
Sectional View



Exploded View

SERVO PARTS





Servo motors are constructed out of basic DC motors, by adding

- Gearreduction
- Positionsensor for themotor shaft
- Electronic circuit that controls the motor's operation

SERVO CONNECTOR

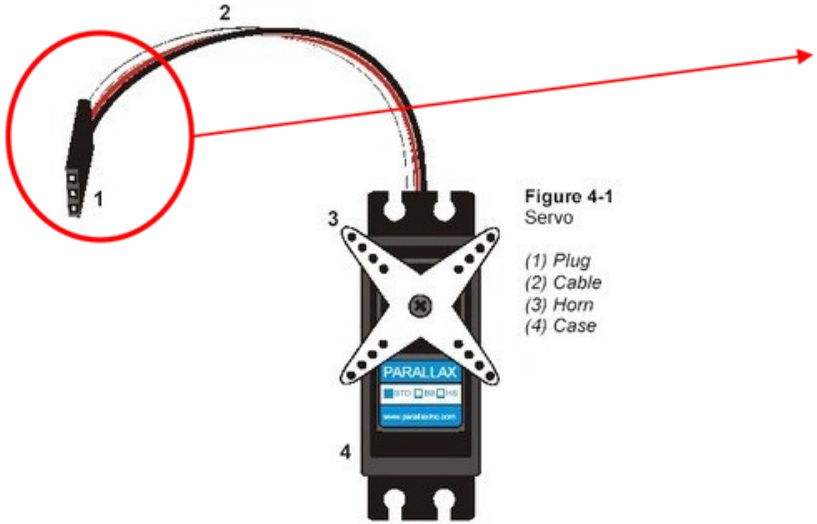
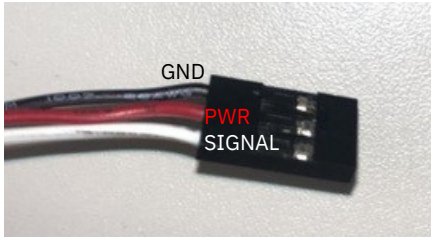
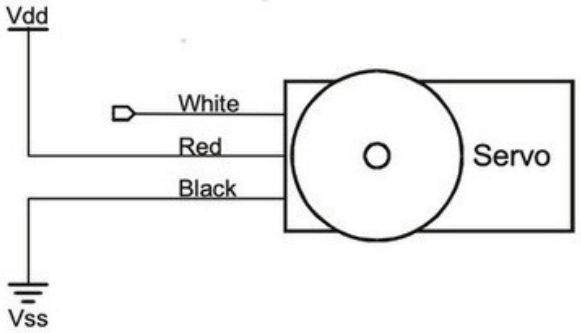


Figure 4-1
Servo

(1) Plug
(2) Cable
(3) Horn
(4) Case

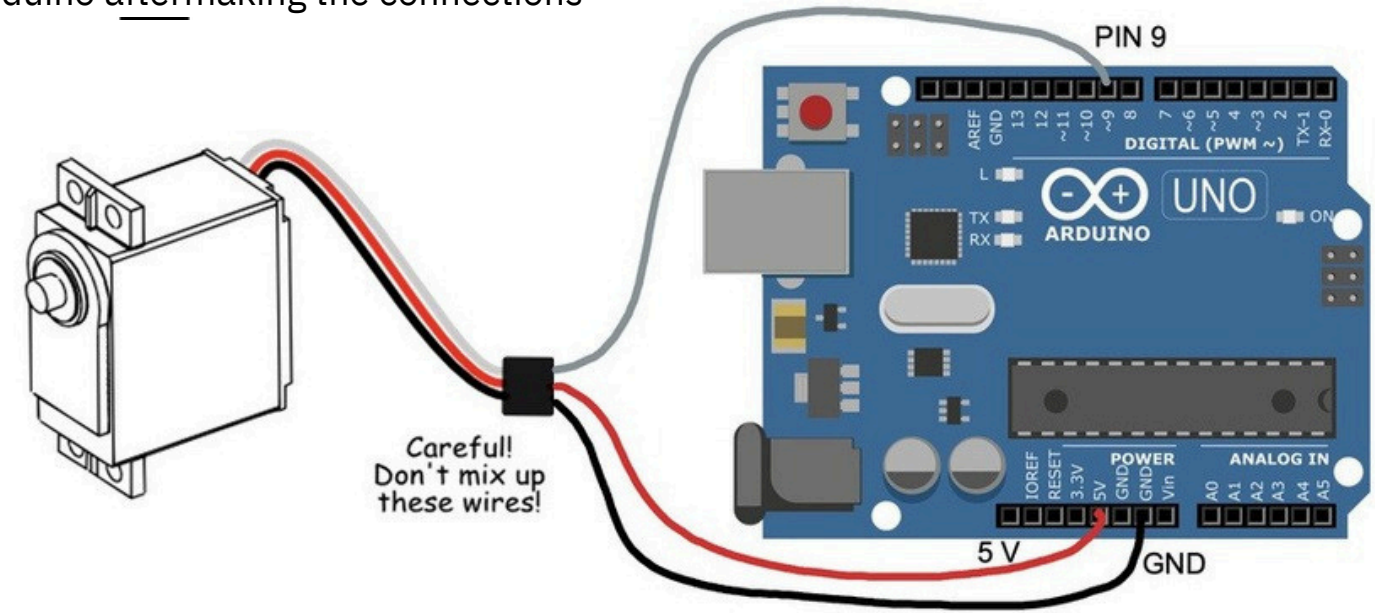


Servo Connector:
Black –ground
Red – power
White –signal



INTERFACING ARDUINO WITH SERVO

NOTE: Always power the Arduino aftermaking the connections



NOTE: Pins with ”~” sign next to the pin number arePWM pins used to control actuators using PWM (here, pin 9 is used to control the servo)

ACTIVITY -3 (SERVO 0°TO 180°)

```
/*Code to rotate servo from 0 to 180°
and back to 0° in steps */

#include <Servo.h>
Servo myservo;
//create servo object to control a servo
int pos=0;
// variable to store the servo position

void setup(){
  myservo.attach(9);
  // attaches the servo on pin 9 to the servo object
}
```

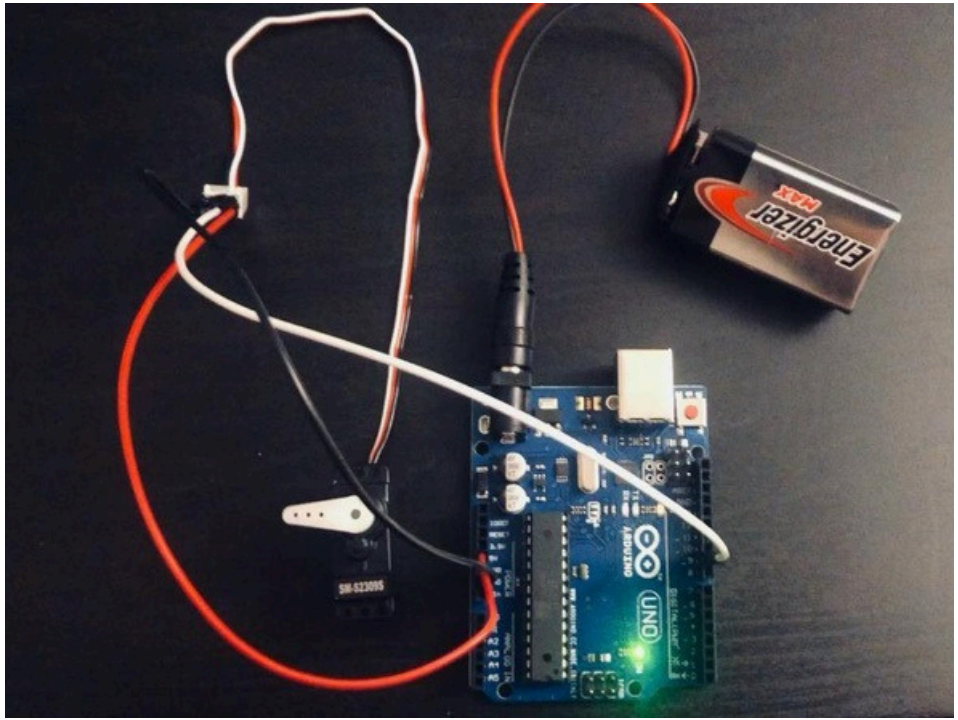
```
void loop(){
  for(pos=0; pos<=180; pos+=1){
    // goes from 0 to 180° in steps of 1°
    myservo.write(pos);
    // tell servo to go to position in variable 'pos'
    delay(15);
    // waits 15ms for the servo to reach the position
  }

  for(pos=180; pos>=0; pos-=1){
    // goes from 180° to 0°
    myservo.write(pos);
    // tell servo to go to position in variable 'pos'
    delay(15);
    // waits 15ms for the servo to reach the position
  }

}
```

Program

ACTIVITY -3 (SERVO 0°TO 180°)



Video

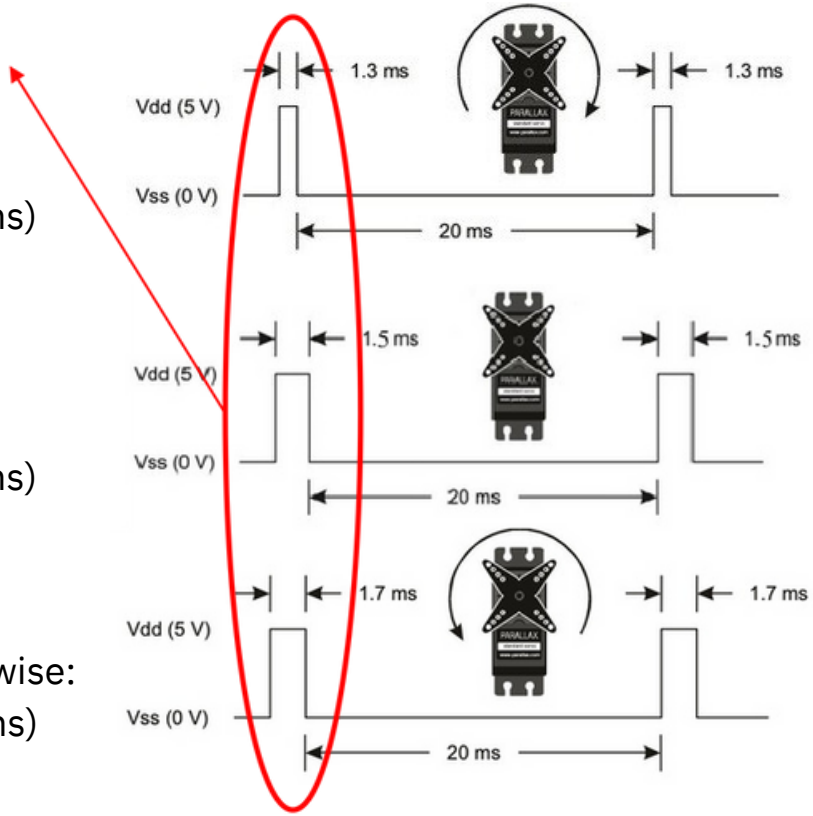
CONTINUOUS SERVO

Pulse length changes to control servo direction

Clockwise:
1300 µs (1.3 ms)

Stopped:
1500 µs (1.5 ms)

Counter-clockwise:
1700 µs (1.7 ms)



- Each pulse is from 1300 to 1700µs (microseconds) in duration
- The pulses repeatabout 50 times each second---once every 20 milliseconds

MOTOR CONSIDERATIONS: SPEED

- More speed requires more voltage

High speed motor	Ordinary motor
Voltage requirement: 6 -8 VDC	Voltage requirement: 4 -6 VDC
Speed: up to 180 RPM	Speed: up to 50 RPM

Speed: (For same power input)



Video High speed servo v/s Ordrinary servo

SPEED ⇑ ⟹ VOLTAGE ⇑

MOTOR CONSIDERATIONS: TORQUE

- More torque requires more current

High speed motor	*Ordinary motor*
Current requirement: 15 -180 mA	Current requirement: 15 -200 mA
Torque: 1.6 +/-0.8 kg-cm @ 7.4 V	Torque: 2.74 kg-cm @ 6 V

Torque: (For same power input)





High Speed Servo v/s Ordinary Servo (Higher Torque)

Video *Video*

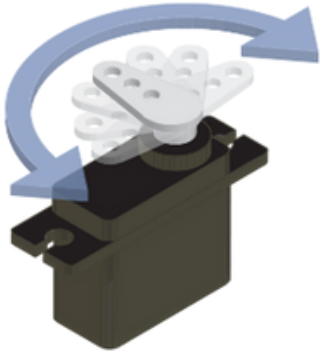
Torque ⇒ Current

CHALLENGE ACTIVITY

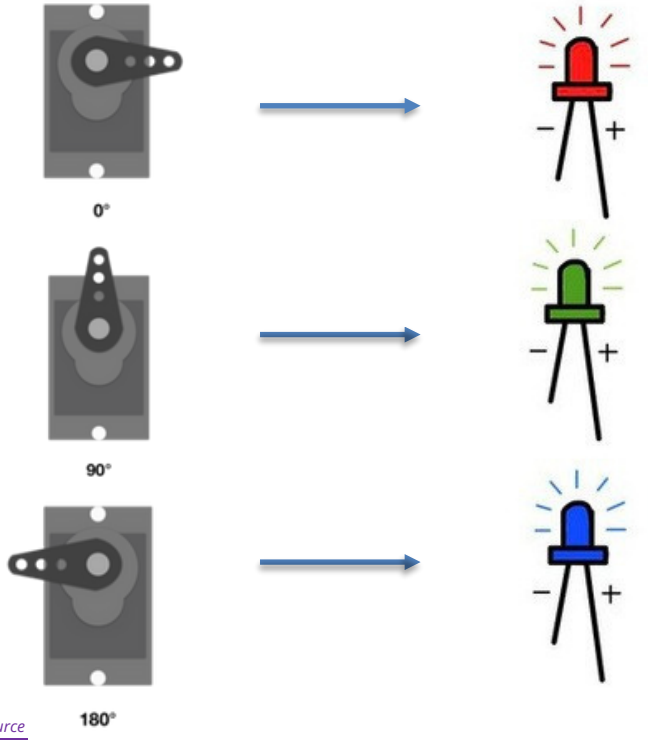
Control servo motor using potentiometer and 3 LEDs to glow at 0, 90 and 180



Source

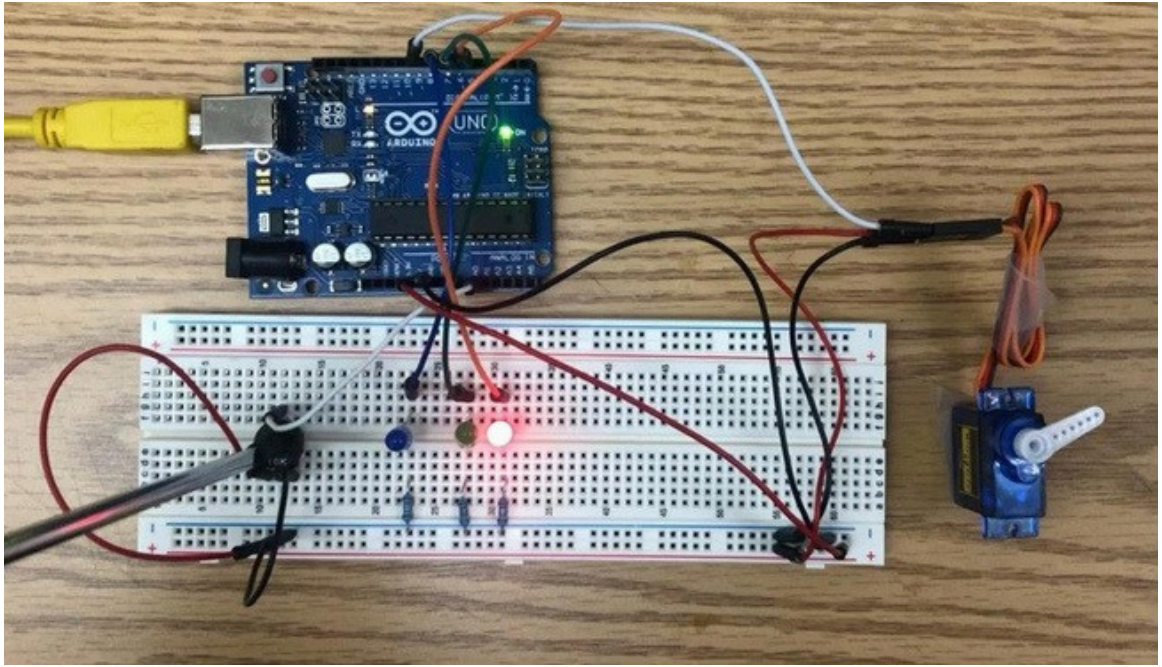


Source



Source

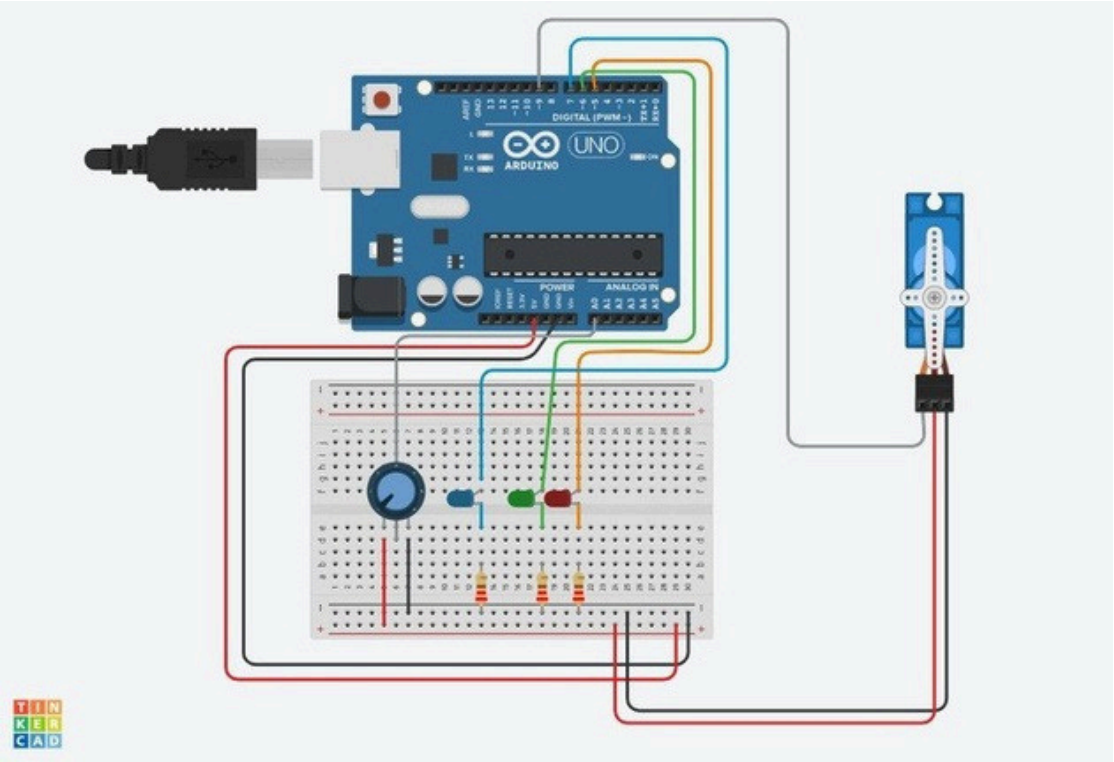
CHALLENGE ACTIVITY -SOLUTION



Video

CHALLENGE ACTIVITY -CIRCUIT

CIRCUIT



CHALLENGE ACTIVITY -CODE

CODE

```
#include <Servo.h>
Servo myservo;
// create servo object to control a servo
int pot_pin = 0;
// analog pin for the potentiometer
int val;
// variable to read the value from the analog pin
int red_led = 5;
int green_led = 6;
int blue_led = 7;
// variable declarations for LED pins
```

```
void setup() {
myservo.attach(9);
// attaches the servo on pin 9 to the servo object
pinMode(red_led, OUTPUT);
pinMode(green_led, OUTPUT);
pinMode(blue_led, OUTPUT);
//sets all LED pins to output mode
}
```

Program

(Contd.)

CHALLENGE ACTIVITY -CODE

CODE

```
void loop() {
digitalWrite(red_led, LOW);
digitalWrite(green_led, LOW);
digitalWrite(blue_led, LOW);
delay(1);
// delay in between readings for stability

val = analogRead(pot_pin);
// reads potentiometer value (value between 0 and 1023)
val = map(val, 0, 1023, 0, 180);
// maps analog value --> servo angle (value between 0 and 180)

myservo.write(val);
// sets the servo position according to the scaled value
delay(15);
// waits for the servo to get there
```

```
if(val < 5){
digitalWrite(red_led, HIGH);
// turn the red LED on
delay(1);
// delay for LED to stay on (avoid visible flickering)
}
if (val > 165){
digitalWrite(blue_led, HIGH);
delay(1);
}
if (val > 80 && val < 110){
digitalWrite(green_led, HIGH);
delay(1);
}
}
```

Program

TASK / ACTIVITY

- Find duty cycle and power, when PWM on time is given
- DC motor direction control for two motors
- DC motor speed control for two motors
- DC motor speed and direction control for two motors
- Servo motor angle control (user input)
- Rotational servo motor Calibration
- Rotational servo speed control
- Rotational servo motor direction control

Thank You!

Questions and Feedback?